

Capacitive touch

DETECT

INPUT

What does it do?

A capacitive touch sensor notices when a finger comes close to its metal pad.

It detects the tiny change in electric charge caused by your body, so no moving parts are needed.

Use it anywhere a project needs a hidden, wipe-clean or feather-light button.

Uses



Touch lamp

DETECT
Finger touches the pad

DECIDE
Is the lamp already on?

DO
Toggle the light



Banana piano

DETECT
A banana is touched

DECIDE
Which note should play?

DO
Play the note



Touch pad
The large metal spiral

SIG
Digital output

VCC and GND
3.3 V power

Code

● Raspberry Pi

```
from gpiozero import DigitalInputDevice

touch = DigitalInputDevice(16, pull_up=False)

touched = touch.is_active
```

● Raspberry Pi Pico

```
from picozero import TouchSensor

touch = TouchSensor(22)

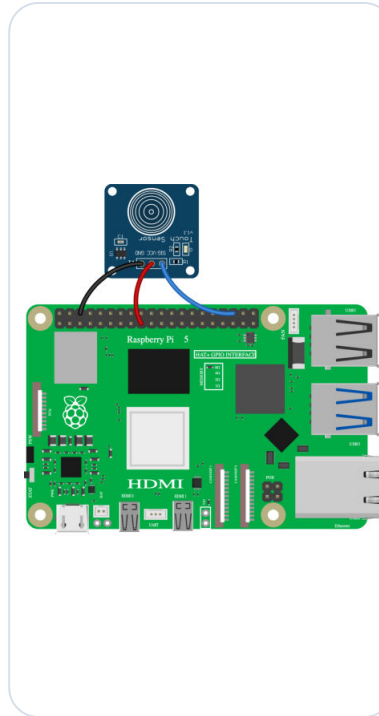
touched = touch.is_touched
```

The TTP223 output goes high while touched.

touched is True or False.

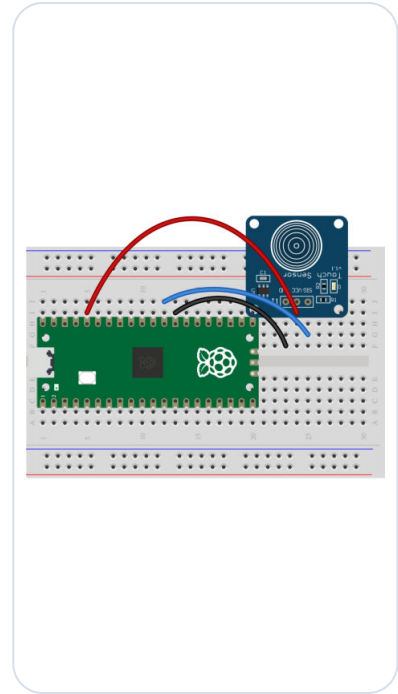
Wiring

RASPBERRY PI 5



GND → GND, **3V3** → VCC, **GPIO 16** → SIG

PICO



GND → GND, **3V3 OUT** → VCC, **GP22** → SIG